



Zearn in Ohio: 2024 Quasi-Experimental Impact Study

Zearn in Ohio

As part of Future Forward Ohio, all Ohio schools and districts have access to Zearn Math for the 2024-25 school year to support math learning acceleration, with Zearn providing access beginning January 2024. Zearn partnered with ODEW to roll out a small implementation pilot for the 2023-24 school year with a subset of schools receiving early access to Zearn. Ohio educators use Zearn digital lessons as a supplement to deepen and reinforce what students are learning with their teacher.

For maximum impact, Zearn recommends students complete 3+ grade-level lessons per week, which is considered fidelity Zearn usage. This generally means 90 minutes-per-week on platform. Zearn also looks at two other levels of consistent usage to capture the incremental effects of Zearn on student achievement. During the pilot rollout in 2023-24, 30K students in G1-8 used Zearn consistently and 6.5K students in G1-8 used Zearn with fidelity (3+ grade-level lessons per week).

Research questions

The Ohio Impact Study draws a sample from the 30K G1-8 Ohio students who used Zearn consistently during the small pilot rollout to analyze two key research questions:

- 1 What is the effect on student achievement on Ohio State Test (OST) for G4-8 students who use Zearn consistently* compared to those who do not?
- 2 What is the effect on student achievement on Ohio State Test (OST) for subgroups of students who use Zearn with fidelity (3+ grade-level lessons per week) compared to those who do not?

Research methodology

This study examines the effect on student achievement on the Ohio State Test (OST) for G4-8 students who use Zearn consistently compared to those who do not use Zearn. Using a treatment-on-the-treated design, researchers compared students who completed Zearn lessons consistently (11K students), completing at least one grade-level lesson each week of the school year, to a matched group of non-users. The treatment group was segmented by usage levels: 1-1.9, 2-2.9, and 3+ grade-level lessons per week (fidelity usage).

The coarsened exact matching method was used to ensure comparable groups. Students were matched based on spring 2023 OST math and ELA scores, grade level, and at least 6 of 9 additional demographic factors to isolate the impact of Zearn usage. Regression models include grade-level dummies to account for grade-level differences in scaling in order to accurately compare results across grade-levels on the OST (a non-vertically scaled assessment).

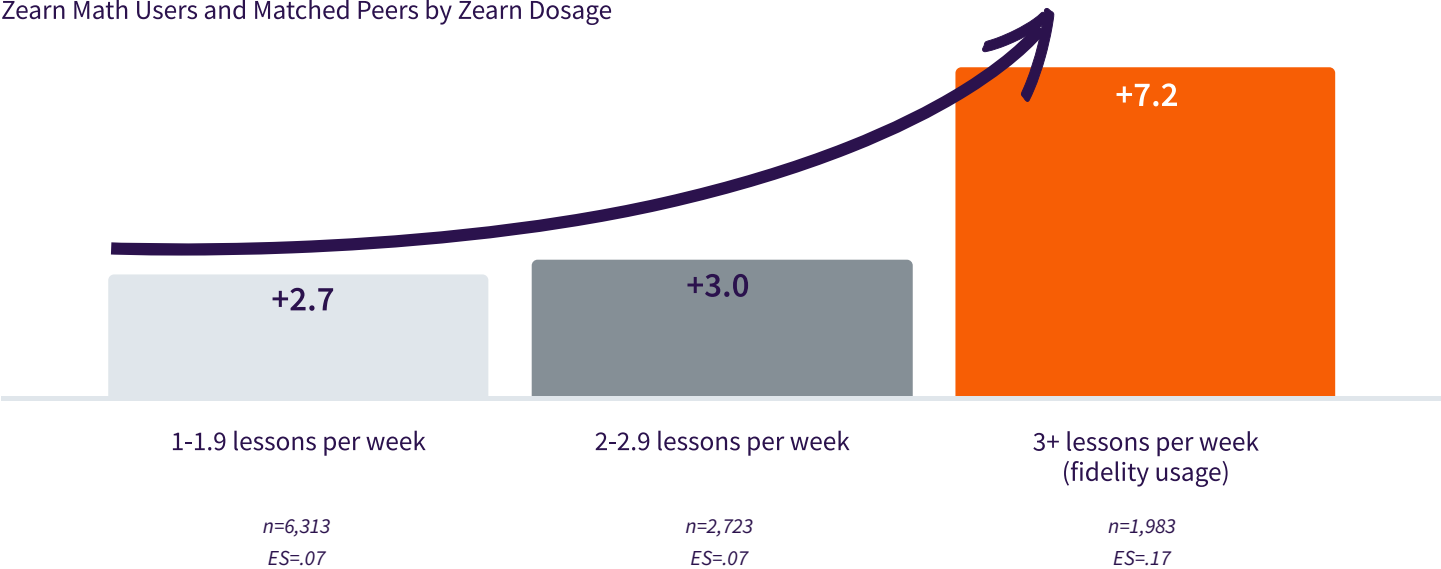
*Defined as completing at least one grade-level lesson per week during the academic school year.

Key findings

KEY FINDING 1

Consistent use of Zearn led to significant positive differences in OST math scores between consistent Zearn users and matched non-users, with the highest impacts (+7.2 scale score points) with fidelity usage (3+ grade-level lessons per week)

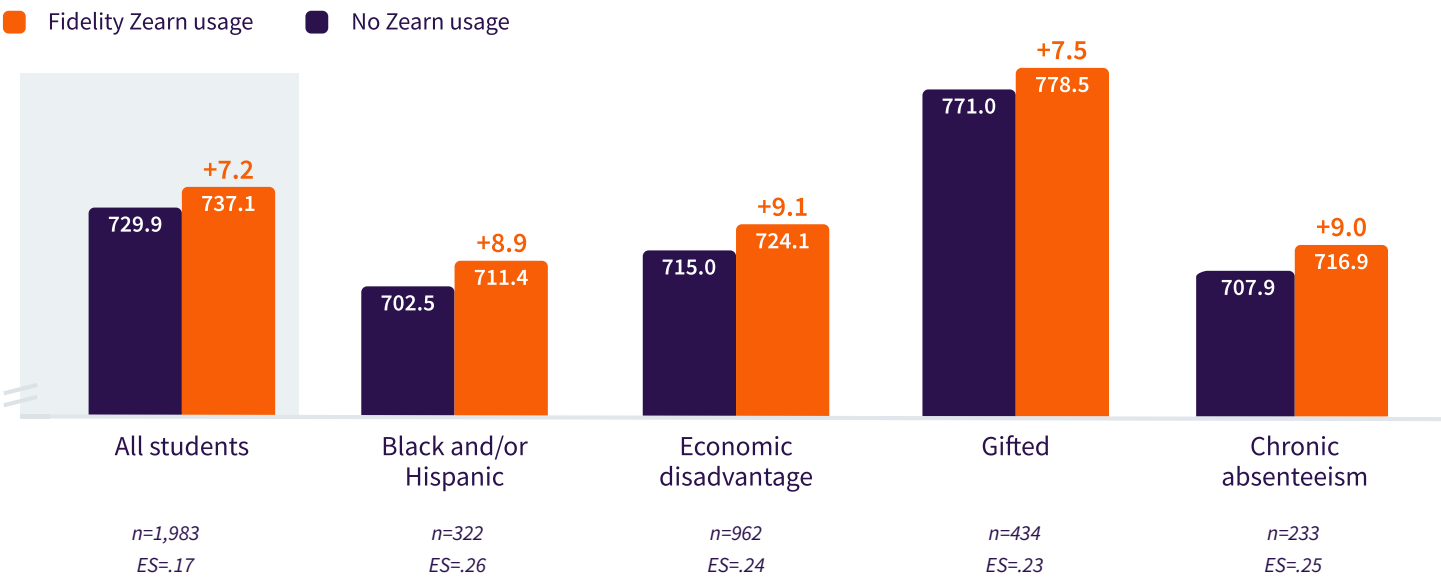
Difference in EOY Scale Score Points on 2024 OST Between Consistent Zearn Math Users and Matched Peers by Zearn Dosage



KEY FINDING 2

Fidelity Zearn usage (3+ grade-level lessons per week) resulted in students scoring 7.2 points higher than matched non-users, with notable gains across sub-groups

Impact of Zearn Math on OST Scores by Subgroup

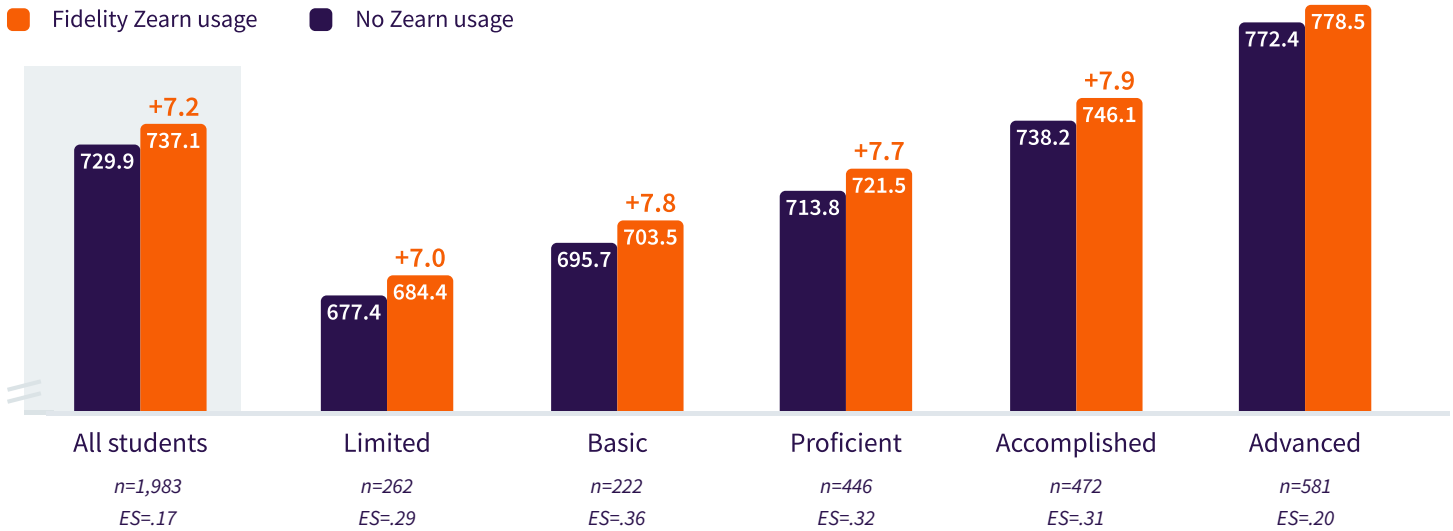


Effect sizes reported are Hedges' g. Subgroups with fewer than 150 students are excluded to ensure robustness of results. All results are statistically significant.

KEY FINDING 3

Fidelity Zearn usage (3+ grade-level lessons per week) resulted in higher scale scores than matched non-users across all starting OST Performance Levels

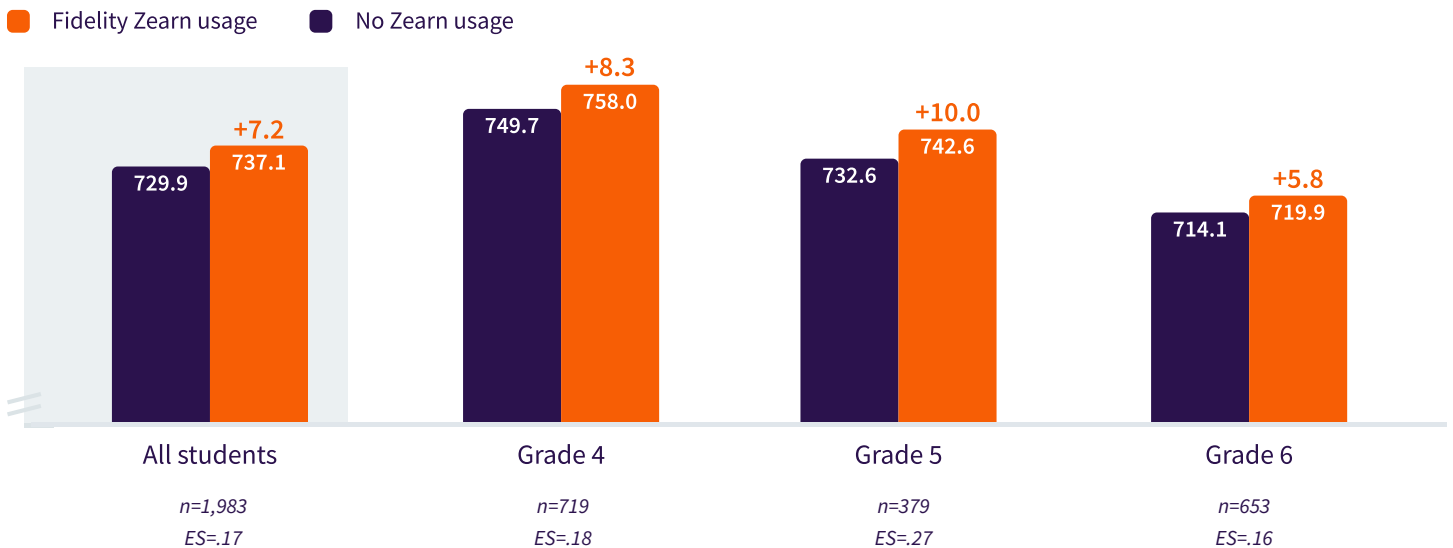
Impact of Zearn Math on OST Scores by Starting Performance Level



KEY FINDING 4

Fidelity Zearn usage (3+ grade-level lessons per week) resulted in higher scale scores than matched non-users across grade levels*

Impact of Zearn Math on OST Scores, by Grade Level



*Grades 7 and 8 were excluded from disaggregation due to samples smaller than 150 students.

Effect sizes reported are Hedges' g. Subgroups with fewer than 150 students are excluded to ensure robustness of results. All results are statistically significant.

Regression analysis detail

- Analysis is based on a regression model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$ in which Y represents 2024 OST math scores, while $\beta_1 \dots \beta_n$ represent student-level characteristics, including treatment or control, 2023 OST math and ELA scores, grade, gender, race, economic disadvantage, English learner, students with disabilities, gifted, and chronic absenteeism.
- Regression models include grade-level dummies to account for grade-level differences in scaling in order to combine all the students in grades 4-8 in the same analysis, given that the OST is not vertically scaled.
- Effect sizes reported are Hedges' g . Subgroups with fewer than 150 students are excluded to ensure robustness of results.
- More information on Zearn's methodology for treatment-on-the-treated impact analyses at webassets.zearn.org/Implementation/EfficacyResearchMethodology.pdf.